Divya Mehul Rajparia

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Education

Degree	University/Institute	Year	CGPA/Marks(%)
B.Tech in Computer Science and Engineering	IIT Hyderabad	2026	9.39
XII High School (Senior Secondary)	Pace Junior Science College	2022	91.50%
X (Secondary School)	Lokhandwala Foundation School	2020	98.17%

Experience

Machine Learning Research Intern - Lab for Machine Learning and Biomedicine *May 2025 – Present*
University of Southern California, Los Angeles, CA

- **Automated tumor segmentation** for breast ultrasound images (**BUSI, BUS-UCLM**), improving diagnostic efficiency and reducing reliance on manual annotation.
- Tackled domain shift across medical datasets by designing a **CycleGAN-based style transfer** method for semantic alignment, improving **DSC by +11.21%** and **IoU by +15.81%**.
- Built a **modular PyTorch pipeline** with synthetic data generation, training, checkpointing, evaluation, and metric logging for reproducible **HPC (SLURM)** experiments.
- Currently exploring a **federated feature-alignment framework** using a UNet encoder and channel-wise mean/variance exchange for **privacy-preserving multi-institutional training**.

Projects

DES: A Smarter Way to Grow ML Models Over Time (PyTorch, Python) — Preprint *Jan 2025 – Present*
Machine Learning and Vision Lab, IIT Hyderabad

- Proposed **AFCIL** (Assumption-Free Class-Incremental Learning), a setup that performs class-incremental learning **without stored data** and handles imbalanced tasks using pre-trained features.
- Designed **Dynamic Expert Saturation (DES)**: a two-stage expert system that first specializes on head classes then expands to tail classes to reduce catastrophic forgetting.
- Implemented the research codebase in **PyTorch**, integrating a frozen **CLIP** backbone with lightweight MLP adapters for efficient adapter-only training and inference.
- Built efficient data loaders and a **class-aware sampler** to handle long-tailed distributions (CIFAR100LT / ImageNetLT) and ensure balanced mini-batches during training.
- Benchmarked DES against strong baselines, achieving up to **+6.38%** accuracy improvements on CIFAR100LT and ImageNetLT.

Sensor-Based Human Activity Classification (TensorFlow, Python) — GitHub *Jan 2025 – Apr 2025*

- Classified human activities from **PIR sensor** readings using a **dual-input CNN-BiLSTM** that fuses temporal features from 109-step time-series with contextual metadata.
- Compared **LSTM, Transformer, and CNN-BiLSTM** architectures through ablation studies; the hybrid model consistently outperformed all baselines.
- Handled severe **class imbalance** (82/11/7% split) to achieve near-perfect recall on minority classes.
- Achieved **99.67% mean accuracy** and **macro F1 = 0.99** across **5-fold cross-validation**.

RISC-V Assembly Code Simulator (C/C++, Computer Architecture) — GitHub *Aug 2024 – Nov 2024*

- Built an **assembly code simulator** to execute RISC-V programs and emulate **registers, memory, stack, and cache** behavior.
- Developed an **assembler** to convert instructions into **machine code** for simulation.
- Implemented **memory segments**: text segment for machine code and data segment for global variables.
- Designed a **stack system** to track function execution and manage local variables.
- Built a **cache simulation** supporting configurable **associativity, block size, and capacity**.

Extracurriculars

Ideation Manager - Entrepreneurship Cell, IIT Hyderabad *May 2023 – May 2024*
Led ideation and planning for **entrepreneurial events**, secured eminent speakers and organized **networking sessions**.

Head - Vibes (Music Club), IIT Hyderabad *May 2023 – May 2024*
Managed all aspects of **club operations**, such as finances, events, media, and membership inductions.

Tech Stack and Skills

Programming: Python, C++, Java, HTML/CSS, SQL
ML Frameworks: PyTorch, TensorFlow, Scikit-Learn
Data Science Tools: NumPy, Pandas, Matplotlib, Seaborn
Tools & Platforms: Git, Linux, SLURM, Jupyter

Please visit my portfolio website for more details about my work.

I am a US Citizen – No Visa Sponsorship Required (US Work Authorized)